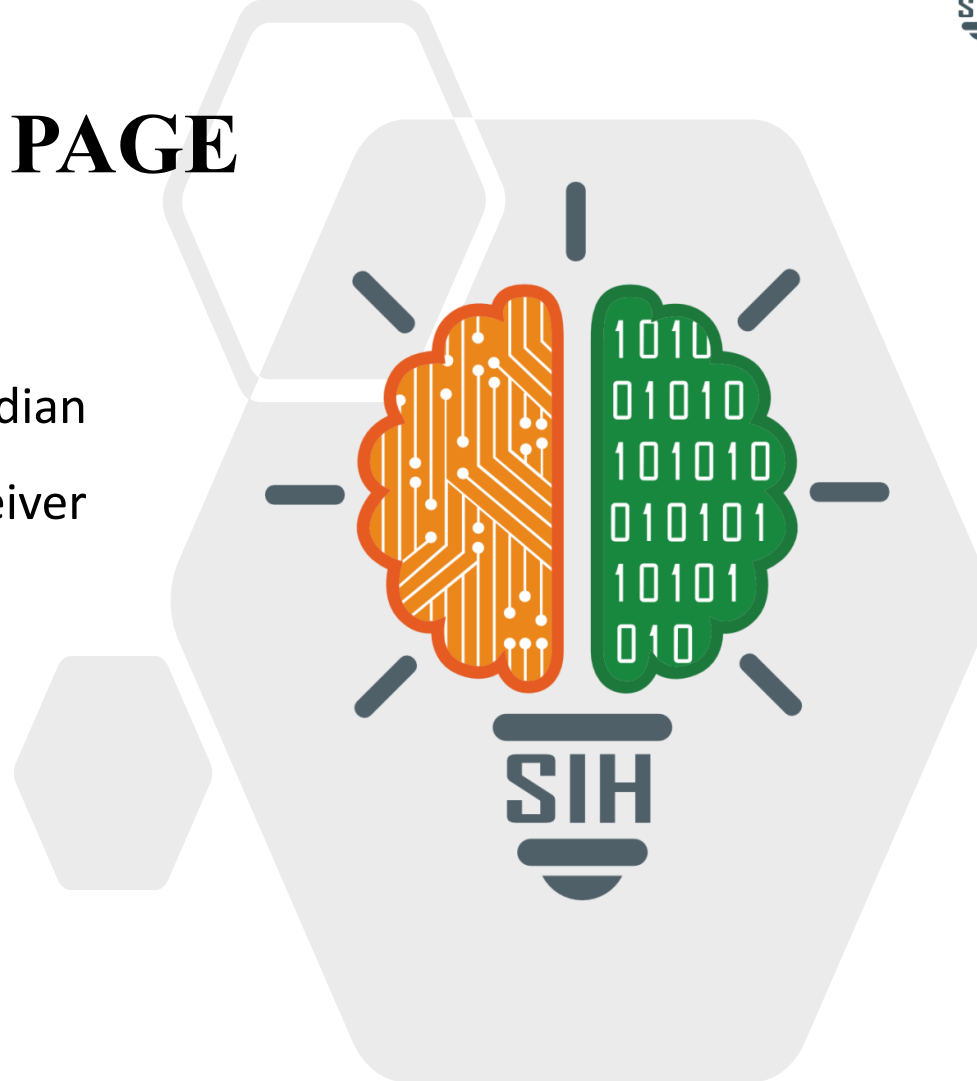


TITLE PAGE

- **Problem Statement ID :- 26173**
- **Problem Statement Title :-** iTantra - Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme :- Smart Automation**
- **PS Category :- Software**
- **Team ID :-**
- **Team Name :- Falcons**



Problem

- ❖ **Raw voice** is highly data-intensive for **low-bitrate** links, while **conventional** STT/TTS often **depends** on **internet** or cloud infrastructure.
- ❖ In **emergencies and remote areas**, this makes fast, inclusive, **multilingual** voice communication **difficult** when **connectivity is unavailable**.

Our Idea

- ❖ **iTantra converts speech into text locally**, transmits only compact data over **Wi-Fi/Bluetooth**, and reconstructs it as **speech** on the **receiving** device.
- ❖ It enables **low-latency, multilingual, offline** voice communication between **phones without** relying on **cellular or cloud infrastructure**.

Unique Value Proposition

- ❖ Transmits **speech as compact text packets** **instead of raw audio**, enabling voice communication over extremely low-bandwidth links.
- ❖ **On-device Edge AI** enables **10+ language communication** with **zero cloud dependency**. During operation, keeping communication private and functional offline.
- ❖ Enables **direct phone-to-phone communication through Wi-Fi Direct with Bluetooth fallback**, with near-instant STT-to-TTS relay for emergency use.

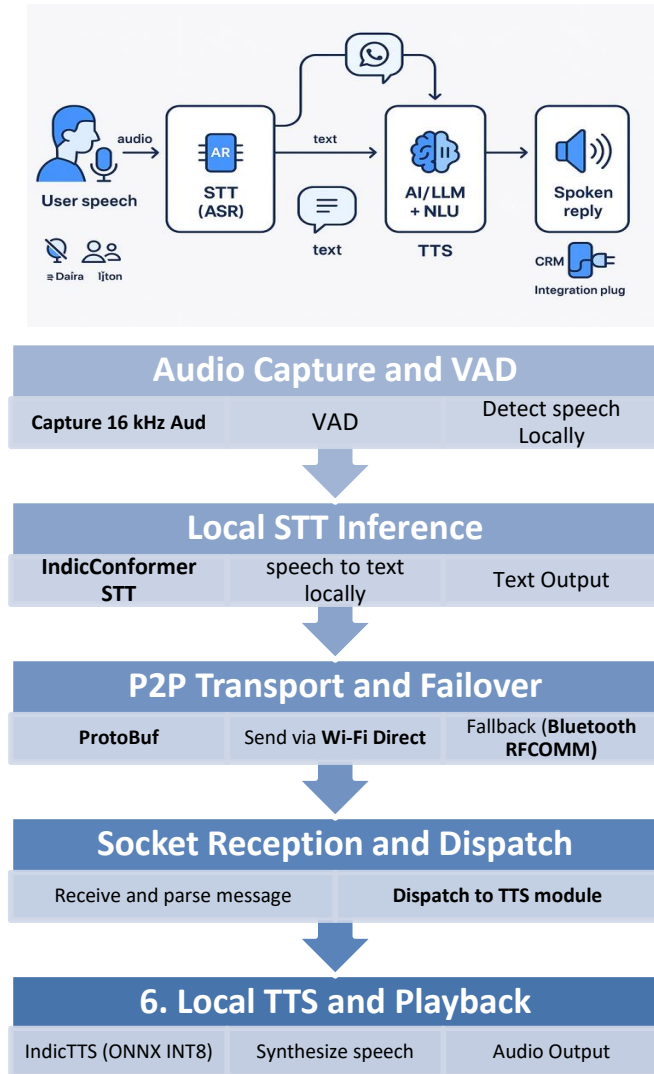


Proposed Solution

- ❖ Captures speech in short chunks and **automatically detects pauses** to finalize utterances.
- ❖ Performs **on-device STT and compact Protobuf packetization**, minimizing the data sent between devices.
- ❖ Uses **Wi-Fi Direct as the primary transport** and **Bluetooth RFCOMM as fallback** for infrastructure-independent communication.

TECHNICAL APPROACH

- ❖ **Edge Ai Runtime:** Voice processing happens **locally** on the device using **ONNX Runtime**, and the **internet** is **only used once** to **download the models**.
- ❖ **Speech Processing:** Captures **16 kHz** mono PCM in **100 ms chunks**, uses **800 ms** pause-aware VAD, with **Indic Conformer STT + VITS TTS** through **sherpa-onnx**.
- ❖ **Data & Communication:** Encodes speech text into compact **Protobuf v3** frames and transmits via **Wi-Fi Direct** / **Wi-Fi P2p Manager**, with **Bluetooth RFCOMM** fallback.
- ❖ **Android Software Stack:** Built with **Kotlin + Jetpack Compose**, using **MVVM/Clean Architecture**, **Coroutines**, **Room**, **DataStore** and **Foreground Service**.



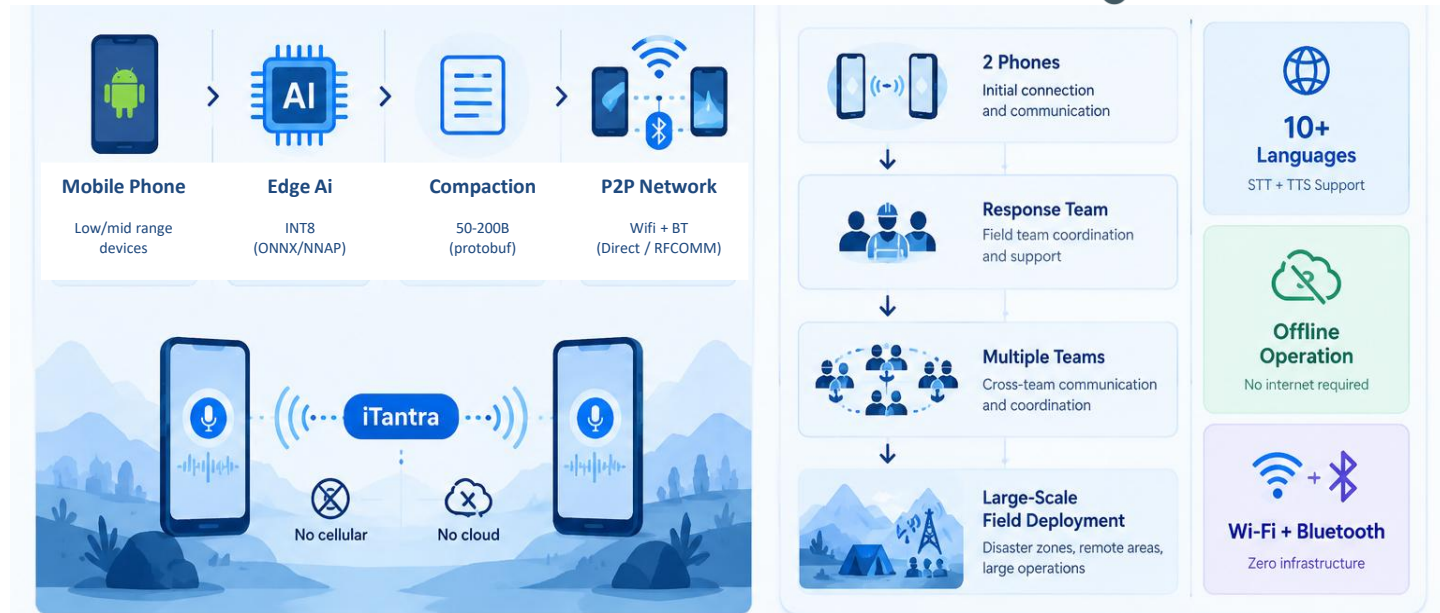
90% Fully Functioning Prototype Provided.
 Demo: <https://youtu.be/SBWvj8v4LYA>
 Our Codebase: <https://github.com/Precise-Goals/iTantra-Communication-Ecosystem>

Feasibility

- **Edge AI Runtime:** Runs locally on mobile chipsets using **ONNX Runtime Mobile and NNAPI**.
- **Storage & Memory:** INT8 models require low storage and **controlled RAM usage**.
- **RF Connectivity:** Uses built-in Wi-Fi Direct and Bluetooth RFCOMM.
- **Open-Source Licensing:** Uses **Apache 2.0** and **MIT-licensed** technologies.

Challenges, Risks & Mitigation

- ❖ Zero Infrastructure Cost: **No cloud, internet and servers** subscriptions required.
- ❖ Power Efficiency: **VAD** reduces STT/TTS processing during silence.
- ❖ Secure **Offline** Design: Operates **without internet** or **cloud** dependency.
- ❖ **Inter-Agency Scalability:** **Protobuf** supports standardized **communication** between teams



Target KPIs

Measured On-Device Benchmark (Helio G99 / 4GB RAM)

Efficiency 20%

Footprint: 215 MB Flash, 262 MB Peak RAM
Idle Load: 48 MB RAM, 2.8% CPU, 1.8% battery/hr

Accuracy 40%

STT (WER): 13.6% average in 15dB field noise
TTS (MOS): 4.10 – 4.40 / 5.0 rating (High legibility)

Latency 20%

End-to-End: 1.85s total (375ms post-speech delivery)
Speed (RTF): 0.25x (Processes 4.5x faster than speech)



Benefits of the solution

01



Instant Deployment

Works without cellular, satellite, or Wi-Fi infrastructure.

02



Bandwidth Reduction

Reduces communication data requirements by over 98%.

03



Low Latency

Provides speech communication with under 2.5-second latency.

01



Multilingual Interoperability

Enables communication across different regional languages.

02



Critical Alert Delivery

Ensures priority SOS alerts receive immediate playback.

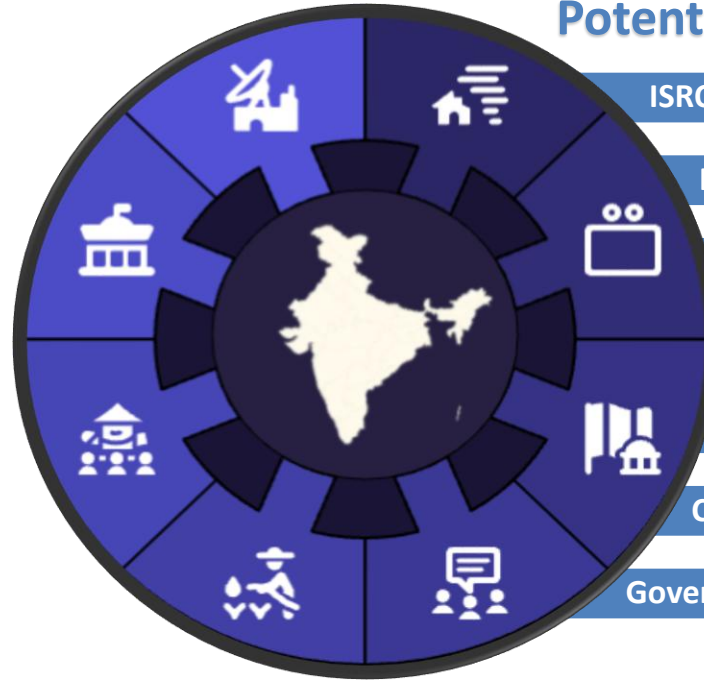
03



Reduced Cognitive Load

Converts distorted speech into clear synthesized audio.

Potential Impact of our Solution



ISRO: Shadow zone telemetry.

First Responders: Zero-infrastructure coordination.

Defense: Air-gapped operational security.

Nation: Reduces import dependency.

Task Forces: Eliminates language friction.

Communities: Stronger local trust.

Government: Simplified live reporting.

Future Scope

- ❖ **Deep-Space Telemetry & LoRa Expansion:** Compressing voice into tiny **50-byte** payloads positions **iTantra** perfectly for **ISRO's** satellite networks, while future **LoRa** hardware integration will extend our **secure, offline terrestrial** mesh beyond **10 kms**.
- ❖ **Accelerating India's Rural Economy: Transforming** everyday smartphones into **decentralized** communication hubs bypasses **vulnerable cellular grids**, ensuring **remote** farmers remain **permanently connected** to drive India's **national growth**.

Semantic Communication & Offline Edge AI:- *Associated Academic Study:* Real-Time **Speech-to-Text** on Edge: A Prototype System for **Ultra-Low** Latency and Local Processing - [MDPI Article](#)

Resilient Ad-Hoc Networking

- ❖ the Framework for **Satellite** Communication Network – [TRAI Paper](#)
- ❖ *Associated Academic Study:* IEEE **Device-to-Device** Wireless **Protocols** – [IEEE Published Research Paper](#)
- ❖ Low Bitrate **High-Quality RVQGAN-based Discrete** Speech Tokenizer – [IBM Research](#)
- ❖ Performance evaluation of **low-bitrate voice** using spread spectrum techniques for satellite-based emergency communication – [Science Direct Article](#)
- ❖ *Associated Academic Study:* An **Infrastructure-Less** Communication Platform for Android Smartphones Using **Wi-Fi Direct** – [Springer Natural Article](#)
- ❖ *Associated Academic Study:* Wi-Fi Direct in Android: Creating seamless **device-to-device** communication – [ArdaScience Article](#)
- ❖ Our **Patent Research** draft: iTantra Universal Communicator – [iTantra Article](#)

Project Ecosystem & Repositories

- ❖ Our Own Codebase: [Our Github Project Repository](#)
- ❖ Deployed Application Link: [Application Download URL](#) & [Application Release Page](#)
- ❖ Video Demonstration: [Prototype Youtube Video Link](#)
- ❖ Live Web Portal: [Landing Page for the Application](#)